From language-model activations to pattern geometry

What is extracted, how a layer is selected, and what enters the comparison with human EEG

1 Present one task trial

Instructions + one solved example

Target sequence

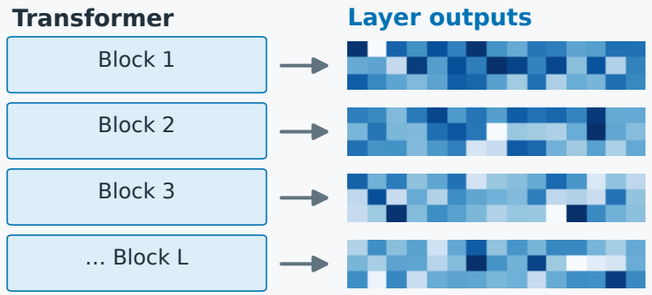
star star star guitar star star star ?

Options: truck / star / cube / guitar

Tokenize the full prompt

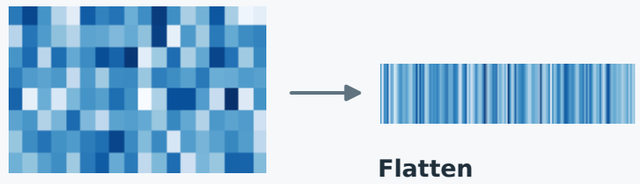
400 independent trials per model
No previous-trial context

2 Read internal activations



Retain the target-sequence token span
Hidden features at every layer

3 Build trial representations



One vector per trial, per layer

Keep token-position information
Stack trial vectors for each layer

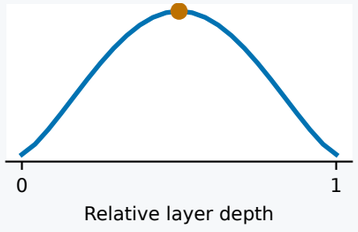
Repeat across layers → choose a layer → compare pattern geometry

4 Select a task-informative layer

Task-design RDM

Same pattern: 0
Different pattern: 1

RSA at each layer

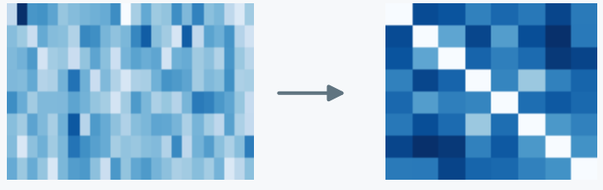


Correlate 400 × 400 trial RDMs with the task-design RDM; retain the best layer.

Layer selection uses no human EEG.

5 Construct the pattern RDM

8 pattern-mean vectors

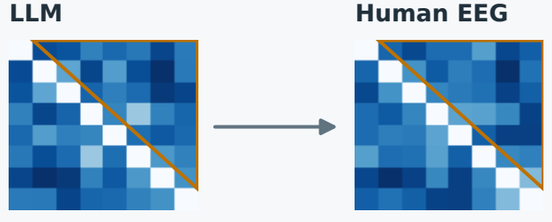


8 × 8 RDM

Average trial vectors within each pattern at the selected layer.

Dissimilarity = 1 − Pearson r
Repeat separately for each model.

6 Compare with human geometry



Extract the same 28 unique pattern pairs

Pearson correlation between RDM entries

One RSA value per model, EEG measure and regional group.

Answers are scored separately for behavioral accuracy; the RSA features are internal activations.

Schematic illustration: token grids, matrices and layer trajectory are not measured data. Eight models are analyzed separately.